

Use Case Description Document

Pet Care Platform — All 35 Use Cases

A. Clinical & Medical | B. Marketplace & Diet | C. Service Provider & Logistics | D. System Administration & Security

A - Clinical & Medical Management

Use Case: Automated Vaccination Scheduler		
Use case name:	Automated Vaccination Scheduler	
Scenario:	Pet owner scheduling vaccinations for their pet	
Triggering event:	Pet is due for a vaccination based on age, species, or medical history	
Brief description:	The system analyzes the pet's profile, vaccination history, and applicable medical protocols to automatically generate and send a vaccination schedule, then confirm the appointment.	
Actors:	Pet Owner, Veterinarian, System Scheduler	
Related use cases:	Chronic Condition Tracker, Digital Prescription Engine, Symptom-to-Specialist Triage	
Stakeholders:	Pet Owner, Veterinary Clinic, Accounting Department	
Preconditions:	Pet profile must exist. Pet's vaccination history must be recorded. A veterinarian must be available.	
Postconditions:	Vaccination appointment is scheduled and saved. Owner receives a confirmation notification. Schedule is updated in the pet's medical record.	
Flow of activities	Actor	System
	1. Owner logs in and views pet profile	1.1 System retrieves pet vaccination history
	2. Owner selects 'Schedule Vaccination'	2.1 System generates vaccination schedule based on pet age and medical protocols
	3. Owner confirms suggested appointment date	3.1 System confirms appointment and sends notification to owner and vet

Exception conditions:	1.1 No available vet slots — system prompts owner to choose alternative date. 2.1 Pet allergy detected for vaccine type — system alerts vet before confirming.
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Use Case: Digital Prescription Engine		
Use case name:	Digital Prescription Engine	
Scenario:	Veterinarian issuing a digital prescription for a pet	
Triggering event:	Veterinarian completes a consultation and determines medication is required	
Brief description:	The veterinarian enters prescription details into the system, which validates dosage against pet weight and species, generates a digital prescription, and notifies the pharmacy or marketplace.	
Actors:	Veterinarian, Pet Owner, System	
Related use cases:	Prescription-Only Purchase Logic, Lab Result Interpretation Hub, Chronic Condition Tracker	
Stakeholders:	Pet Owner, Veterinary Clinic, Pharmacy/Marketplace, Accounting	
Preconditions:	Consultation record must exist. Pet profile and medical history must be available. Veterinarian must be licensed and authenticated.	
Postconditions:	Digital prescription is generated and stored. Pet owner is notified. Marketplace is authorized to fulfill prescription order.	
Flow of activities	Actor	System
	1. Vet selects pet record after consultation	1.1 System loads pet medical history and allergies
	2. Vet enters medication, dosage, and duration	2.1 System validates dosage against pet weight and species
	3. Vet submits prescription	3.1 System generates digital prescription and notifies pharmacy and owner
Exception conditions:	2.1 Dosage exceeds safe limit — system blocks submission and shows warning. 3.1 Pharmacy integration failure — system queues prescription and retries.	

Use Case: Symptom-to-Specialist Triage

Use case name:	Symptom-to-Specialist Triage	
Scenario:	Pet owner reporting symptoms and seeking the right specialist	
Triggering event:	Pet owner inputs symptoms through the platform's symptom checker	
Brief description:	The system processes reported symptoms using a medical decision engine to classify urgency and recommend the most appropriate specialist type, providing guidance on next steps.	
Actors:	Pet Owner, System Triage Engine, Veterinary Specialist	
Related use cases:	Emergency Red-Flag Logic, Automated Vaccination Scheduler, Lab Result Interpretation Hub	
Stakeholders:	Pet Owner, Veterinary Specialists, Platform Administration	
Preconditions:	Pet profile must be available. Symptom checker module must be active. Specialist database must be populated.	
Postconditions:	Triage result is recorded. Recommended specialist is identified. Owner receives guidance and optional appointment link.	
Flow of activities	Actor	System
	1. Owner navigates to symptom checker	1.1 System loads symptom checker interface
	2. Owner inputs symptoms and duration	2.1 System processes symptoms through medical decision engine
	3. Owner reviews triage result and selects specialist	3.1 System lists suitable specialists and allows booking
Exception conditions:	2.1 Symptoms match emergency criteria — system immediately triggers Emergency Red-Flag Logic. 3.1 No available specialist — system suggests nearest clinic or emergency contact.	

Use Case: Chronic Condition Tracker	
Use case name:	Chronic Condition Tracker
Scenario:	Veterinarian or pet owner monitoring a pet's ongoing chronic condition
Triggering event:	Pet is diagnosed with a chronic condition requiring long-term monitoring
Brief description:	The system enables ongoing tracking of vital health parameters for pets with chronic conditions, sending alerts when readings are abnormal and generating trend reports for veterinary review.

Actors:	Pet Owner, Veterinarian, System	
Related use cases:	Digital Prescription Engine, Lab Result Interpretation Hub, Weight & Nutrition Trend Graphing	
Stakeholders:	Pet Owner, Veterinary Clinic, Health Analytics Team	
Preconditions:	Chronic condition diagnosis must be recorded. Monitoring parameters must be configured by the vet. Pet profile must be active.	
Postconditions:	Health data is logged and stored. Alerts are sent if abnormal readings detected. Trend report is generated for vet review.	
Flow of activities	Actor	System
	1. Owner logs daily health readings (weight, glucose, etc.)	1.1 System validates and stores readings
	2. Vet reviews trend report on dashboard	2.1 System generates visual trend report
	3. Vet updates monitoring parameters if needed	3.1 System applies new monitoring thresholds and notifies owner
Exception conditions:	1.1 Reading is critically abnormal — system triggers immediate alert to vet and owner. 2.1 Insufficient data for trend — system prompts owner to complete missing entries.	

Use Case: Surgery & Procedure Resource Manager	
Use case name:	Surgery & Procedure Resource Manager
Scenario:	Clinic administrator scheduling a surgical procedure
Triggering event:	Veterinarian recommends surgery for a pet following consultation
Brief description:	The system manages the allocation of surgical rooms, equipment, and veterinary staff for scheduled procedures, preventing resource conflicts and ensuring all requirements are met before the surgery date.
Actors:	Veterinary Surgeon, Clinic Administrator, System
Related use cases:	Sitter Availability Conflict Resolver, Provider Certification Verification
Stakeholders:	Veterinary Clinic, Pet Owner, Accounting Department

Preconditions:	Surgical procedure must be recommended and approved. Clinic resources must be registered in the system. Veterinary surgeon must be certified.	
Postconditions:	Surgery is scheduled with allocated resources. Owner receives pre-surgery instructions. Resource calendar is updated.	
Flow of activities	Actor	System
	1. Admin selects procedure type and preferred date	1.1 System checks availability of rooms, equipment, and surgeon
	2. Admin confirms resource allocation	2.1 System reserves resources and blocks calendar
	3. Admin sends pre-surgery instructions to owner	3.1 System sends automated pre-surgery checklist to owner
Exception conditions:	1.1 Required equipment unavailable on chosen date — system suggests alternative dates. 2.1 Surgeon not certified for procedure type — system blocks allocation and alerts admin.	

Use Case: Lab Result Interpretation Hub		
Use case name:	Lab Result Interpretation Hub	
Scenario:	Veterinarian reviewing laboratory test results for a pet	
Triggering event:	Lab results are uploaded or received from an external laboratory	
Brief description:	The system receives lab results, compares values to species-specific normal ranges, flags abnormal findings, and generates an interpretive summary report for the veterinarian.	
Actors:	Veterinarian, Laboratory System, Pet Owner	
Related use cases:	Chronic Condition Tracker, Digital Prescription Engine, Symptom-to-Specialist Triage	
Stakeholders:	Veterinary Clinic, Pet Owner, Laboratory Partner	
Preconditions:	Lab order must be placed and linked to pet record. Laboratory system integration must be active. Normal range database must be updated.	
Postconditions:	Results are stored in pet medical record. Abnormal findings are flagged. Interpretive report is generated and shared with vet and owner.	
Flow of activities	Actor	System

	1. Vet orders lab test through platform	1.1 System sends lab order to integrated laboratory
	2. Lab technician uploads results	2.1 System parses results and compares to species-specific normal ranges
	3. Vet reviews interpretive report	3.1 System flags abnormal values and generates summary report
Exception conditions:	2.1 Result format not recognized — system alerts lab to re-upload in correct format. 3.1 Critical value detected — system immediately notifies vet and owner.	

Use Case: Veterinary Referrals Workflow		
Use case name:	Veterinary Referrals Workflow	
Scenario:	Veterinarian referring a pet to a specialist clinic	
Triggering event:	Primary vet determines the case requires specialist attention	
Brief description:	The system facilitates the formal referral process, transferring relevant medical records, appointment details, and communication between the referring vet and specialist clinic.	
Actors:	Primary Veterinarian, Specialist Veterinarian, Pet Owner, System	
Related use cases:	Symptom-to-Specialist Triage, Lab Result Interpretation Hub, Behavioral Profile Sharing	
Stakeholders:	Referring Clinic, Specialist Clinic, Pet Owner	
Preconditions:	Pet medical record must be complete and up to date. Specialist clinic must be registered on the platform. Owner must provide consent for record transfer.	
Postconditions:	Referral is sent to specialist. Medical records are securely transferred. Owner receives confirmation and specialist appointment details.	
Flow of activities	Actor	System
	1. Vet initiates referral from pet record	1.1 System prepares referral package with medical history
	2. Vet selects appropriate specialist and clinic	2.1 System checks specialist availability and sends referral request

	3. Owner approves record sharing consent	3.1 System transfers encrypted records and confirms appointment
Exception conditions:	2.1 Specialist clinic not available — system lists alternative specialists. 3.1 Owner denies consent — system halts record transfer and notifies vet.	

Use Case: Weight & Nutrition Trend Graphing		
Use case name:	Weight & Nutrition Trend Graphing	
Scenario:	Pet owner or veterinarian tracking a pet's weight and nutritional intake over time	
Triggering event:	Owner logs a new weight or nutrition entry, or vet requests a trend report	
Brief description:	The system records weight and nutritional data entries, generates visual trend graphs, compares values to breed and age benchmarks, and alerts the vet or owner when trends indicate concern.	
Actors:	Pet Owner, Veterinarian, System	
Related use cases:	Chronic Condition Tracker, Medical-Diet Compatibility Engine, Digital Prescription Engine	
Stakeholders:	Pet Owner, Veterinary Clinic, Health Analytics Team	
Preconditions:	Pet profile with breed, age, and species must exist. Weight and nutrition logging module must be active. Benchmark data must be configured in the system.	
Postconditions:	New data entry is saved. Trend graph is updated and available to owner and vet. Alert is triggered if values fall outside healthy range.	
Flow of activities	Actor	System
	1. Owner logs pet weight and daily food intake	1.1 System stores entry and updates trend graph
	2. Owner or vet views trend graph	2.1 System displays weight and nutrition trend with benchmark overlay
	3. Vet reviews and adds dietary recommendation if needed	3.1 System saves dietary recommendation and links it to pet profile
Exception conditions:	1.1 Entry value is outside plausible range — system prompts owner to confirm or correct input.	

	3.1 No benchmark data available for breed/age — system displays raw trend without benchmark comparison.
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Use Case: Emergency Red-Flag Logic		
Use case name:	Emergency Red-Flag Logic	
Scenario:	System detecting a life-threatening situation for a pet	
Triggering event:	Symptom input or health reading triggers a critical emergency threshold	
Brief description:	The system automatically identifies emergency red-flag conditions based on symptom combinations or abnormal vitals, and immediately escalates the case with urgent alerts to the owner, vet, and emergency services if needed.	
Actors:	System Emergency Engine, Pet Owner, Emergency Veterinarian	
Related use cases:	Symptom-to-Specialist Triage, Chronic Condition Tracker, Multi-Channel Notification Escalator	
Stakeholders:	Pet Owner, Emergency Veterinary Service, Platform Administration	
Preconditions:	Emergency red-flag rules must be defined and active. Pet profile must be available. Notification channels must be configured.	
Postconditions:	Emergency alert is sent to all relevant parties. Case is escalated as urgent. Emergency contact and nearest clinic details are provided to owner.	
Flow of activities	Actor	System
	1. Owner inputs symptoms or system receives critical health reading	1.1 System applies emergency rule engine to inputs
	2. Owner receives emergency alert with guidance	2.1 System identifies red-flag condition and triggers emergency alert
		2.2 System provides nearest emergency clinic details and first-aid guidance
Exception conditions:	1.1 False positive detected — system includes option for owner to dismiss alert and log reason. 2.1 Notification delivery failure — system retries via alternative channel.	

B - Marketplace & Diet Integration

Use Case: Medical-Diet Compatibility Engine		
Use case name:	Medical-Diet Compatibility Engine	
Scenario:	Pet owner browsing diet products for a pet with medical conditions	
Triggering event:	Owner searches for food or supplement products for a pet with an existing medical condition	
Brief description:	The system cross-references the pet's medical conditions and prescribed medications with available diet products, filtering incompatible items and recommending medically appropriate options.	
Actors:	Pet Owner, Veterinarian, System	
Related use cases:	Ingredient Allergy Alert System, Prescription-Only Purchase Logic, Chronic Condition Tracker	
Stakeholders:	Pet Owner, Veterinary Clinic, Marketplace Vendors	
Preconditions:	Pet medical profile must be active and updated. Product ingredient database must be current. Vet dietary recommendations must be recorded.	
Postconditions:	Incompatible products are filtered from results. Compatible products are recommended. Owner is informed of any dietary restrictions.	
Flow of activities	Actor	System
	1. Owner searches for pet food or supplements	1.1 System retrieves pet medical profile and dietary flags
	2. Owner selects a product to view details	2.1 System checks ingredient compatibility against medical conditions
	3. Owner adds compatible product to cart	3.1 System confirms compatibility and allows purchase
Exception conditions:	2.1 Product ingredient conflicts with pet condition — system displays warning and suggests alternatives. 1.1 Medical profile incomplete — system prompts owner to update pet record before browsing.	

Use Case: Prescription-Only Purchase Logic	
Use case name:	Prescription-Only Purchase Logic

Scenario:	Owner attempting to purchase a prescription medication or diet product	
Triggering event:	Owner adds a prescription-only product to the shopping cart	
Brief description:	The system verifies the existence of a valid prescription linked to the pet before allowing the purchase of prescription-only items, rejecting unauthorized purchases.	
Actors:	Pet Owner, Veterinarian, System	
Related use cases:	Digital Prescription Engine, Medical-Diet Compatibility Engine, Inventory Expiry Tracker	
Stakeholders:	Pet Owner, Veterinary Clinic, Marketplace, Legal/Compliance Team	
Preconditions:	Prescription-only product must be flagged in the product catalog. A valid and active prescription must be on file. Pet profile must be linked to owner account.	
Postconditions:	Purchase is approved if valid prescription exists. Purchase is rejected with explanation if no prescription found. Transaction is logged for compliance.	
Flow of activities	Actor	System
	1. Owner selects prescription-only product	1.1 System flags product as prescription-only
	2. Owner proceeds to checkout	2.1 System checks for valid prescription linked to pet
		2.2 System approves or rejects purchase based on prescription status
Exception conditions:	2.1 Prescription expired — system notifies owner and directs them to request a renewal from vet. 2.2 No prescription on file — system blocks purchase and displays instructions to obtain prescription.	

Use Case: Smart Auto-Ship Subscription	
Use case name:	Smart Auto-Ship Subscription
Scenario:	Owner setting up recurring delivery for pet products
Triggering event:	Owner selects 'Subscribe & Auto-Ship' option on a product page
Brief description:	The system creates a recurring delivery schedule for selected products, automatically processing orders at the defined frequency,

	adjusting quantities based on usage history, and notifying the owner before each shipment.	
Actors:	Pet Owner, System, Logistics Partner	
Related use cases:	Inventory Expiry Tracker, Vet-Recommended Reward Linkage, Loyalty Health Points Program	
Stakeholders:	Pet Owner, Marketplace Vendors, Logistics, Accounting	
Preconditions:	Owner must have an active account with saved payment method. Product must be available for subscription. Owner must set delivery frequency.	
Postconditions:	Subscription is created and scheduled. First order is placed immediately or on specified date. Owner receives subscription confirmation and upcoming schedule.	
Flow of activities	Actor	System
	1. Owner selects product and clicks 'Subscribe'	1.1 System checks product availability for subscription
	2. Owner sets delivery frequency and quantity	2.1 System calculates next delivery dates
	3. Owner confirms subscription	3.1 System creates subscription record, charges payment, and schedules shipment
Exception conditions:	3.1 Payment method declined — system notifies owner to update payment before activating subscription. 2.1 Product discontinued — system alerts owner to select replacement product.	

Use Case: Ingredient Allergy Alert System	
Use case name:	Ingredient Allergy Alert System
Scenario:	Owner browsing or purchasing a product containing a pet allergen
Triggering event:	Owner views or adds to cart a product that contains an ingredient flagged as an allergen for their pet
Brief description:	The system checks product ingredients against the pet's documented allergy profile and immediately alerts the owner if an allergen is detected, preventing accidental purchase of harmful products.
Actors:	Pet Owner, System, Veterinarian
Related use cases:	Medical-Diet Compatibility Engine, Prescription-Only Purchase Logic, Product Recall Notification System

Stakeholders:	Pet Owner, Veterinary Clinic, Marketplace Vendors, Legal/Safety Team	
Preconditions:	Pet allergy profile must be documented and saved. Product ingredient list must be available in the system. Owner must be logged in.	
Postconditions:	Allergen alert is displayed to owner. Purchase of allergen-containing product is flagged or blocked. Incident is logged in pet health record.	
Flow of activities	Actor	System
	1. Owner searches or browses products	1.1 System loads pet allergy profile in background
	2. Owner selects a product for review	2.1 System scans product ingredients against pet allergy profile
		2.2 System displays allergen warning if match found
Exception conditions:	2.1 Ingredient data missing from product — system displays warning that ingredient info is incomplete. 2.2 Owner overrides warning — system logs override and requires confirmation.	

Use Case: Vet-Recommended Reward Linkage	
Use case name:	Vet-Recommended Reward Linkage
Scenario:	Owner purchasing vet-recommended products and earning loyalty rewards
Triggering event:	Owner purchases a product that has been explicitly recommended by their pet's veterinarian
Brief description:	The system links veterinarian product recommendations to the marketplace catalog and applies bonus loyalty points or rewards when owners purchase those recommended products.
Actors:	Pet Owner, Veterinarian, System
Related use cases:	Loyalty Health Points Program, Digital Prescription Engine, Smart Auto-Ship Subscription
Stakeholders:	Pet Owner, Veterinary Clinic, Marketplace, Marketing Team
Preconditions:	Vet must have submitted a product recommendation linked to pet record. Product must be available in the marketplace. Loyalty program must be active.
Postconditions:	Purchase is completed. Bonus loyalty points are credited to owner account. Vet recommendation is marked as fulfilled.

Flow of activities	Actor	System
	1. Owner views vet-recommended products	1.1 System retrieves vet recommendations linked to pet
	2. Owner selects and purchases a recommended product	2.1 System processes purchase and calculates bonus loyalty points
		2.2 System credits points and sends confirmation to owner
Exception conditions:	2.1 Recommended product out of stock — system suggests equivalent alternatives. 1.1 No vet recommendations on file — system prompts owner to request recommendation at next consultation.	

Use Case: Inventory Expiry Tracker		
Use case name:	Inventory Expiry Tracker	
Scenario:	System administrator monitoring product expiry dates in the marketplace inventory	
Triggering event:	Products approach their expiry date threshold	
Brief description:	The system continuously monitors product expiry dates in the inventory, automatically flagging items approaching expiry, removing expired listings, and notifying vendors and admins for action.	
Actors:	System Admin, Marketplace Vendor, System	
Related use cases:	Product Recall Notification System, Prescription-Only Purchase Logic, Multi-Vendor Commission Logic	
Stakeholders:	Marketplace Vendors, Platform Administration, Pet Owners, Regulatory Compliance	
Preconditions:	Product expiry dates must be entered into the system. Expiry threshold rules must be configured by admin. Vendor accounts must be active.	
Postconditions:	Expiring products are flagged and listed for review. Expired products are automatically removed from marketplace. Vendors receive notification for restocking.	
Flow of activities	Actor	System
	1. Admin configures expiry alert threshold (e.g., 30 days)	1.1 System schedules daily expiry scan

	2. Vendor reviews flagged expiring products	2.1 System flags products nearing expiry threshold
	3. Vendor updates or removes product listings	3.1 System auto-removes expired listings and logs action
Exception conditions:	2.1 Vendor fails to act on flagged items before expiry — system automatically removes listing and notifies admin. 1.1 Expiry date missing from product — system marks product as incomplete and blocks listing.	

Use Case: Multi-Vendor Commission Logic		
Use case name:	Multi-Vendor Commission Logic	
Scenario:	Marketplace processing sales and calculating vendor commissions	
Triggering event:	A product sale is completed on the marketplace by a registered vendor	
Brief description:	The system applies the correct commission rate per vendor agreement upon sale completion, calculates the vendor's payable amount, and records the transaction for accounting and payout processing.	
Actors:	Marketplace Vendor, System, Accounting Team	
Related use cases:	Service Escrow Management, Inventory Expiry Tracker, Loyalty Health Points Program	
Stakeholders:	Marketplace Vendors, Accounting/Finance Team, Platform Administration	
Preconditions:	Vendor must be registered with a commission agreement on file. Product must be listed and active. Payment must be successfully processed.	
Postconditions:	Commission is calculated and recorded. Vendor payout amount is updated. Accounting records reflect the transaction.	
Flow of activities	Actor	System
	1. Vendor lists product on marketplace	1.1 System stores product with vendor commission rate
	2. Customer completes purchase	2.1 System calculates platform commission and vendor payout
		2.2 System records transaction and updates vendor balance dashboard

Exception conditions:	2.1 Commission rate agreement missing — system flags transaction for manual review. 2.2 Payment processing failure — system reverses commission calculation and notifies accounting.
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Use Case: Loyalty Health Points Program		
Use case name:	Loyalty Health Points Program	
Scenario:	Owner earning and redeeming health loyalty points through purchases and activities	
Triggering event:	Owner completes a qualifying action (purchase, vet visit, health tracking update)	
Brief description:	The system awards loyalty health points for qualifying activities, maintains a running points balance, and allows owners to redeem points for discounts or rewards at checkout.	
Actors:	Pet Owner, System, Marketplace	
Related use cases:	Vet-Recommended Reward Linkage, Smart Auto-Ship Subscription, Multi-Vendor Commission Logic	
Stakeholders:	Pet Owners, Marketing Team, Accounting, Platform Administration	
Preconditions:	Owner must have an active account enrolled in the loyalty program. Qualifying activities must be defined and configured in the system.	
Postconditions:	Points are credited or deducted accurately. Owner's points balance is updated in real time. Redemption is applied to order total.	
Flow of activities	Actor	System
	1. Owner completes a qualifying action	1.1 System detects qualifying action and credits appropriate points
	2. Owner views points balance	2.1 System displays updated points balance and expiry info
	3. Owner redeems points at checkout	3.1 System deducts redeemed points and applies discount to order
Exception conditions:	3.1 Insufficient points for selected redemption — system shows minimum required and current balance. 1.1 Duplicate action detected — system prevents double-awarding of points.	

Use Case: Product Recall Notification System		
Use case name:	Product Recall Notification System	
Scenario:	Platform notifying affected pet owners of a product recall	
Triggering event:	A product sold on the marketplace is subject to a safety or quality recall	
Brief description:	The system identifies all customers who purchased a recalled product, sends immediate recall notifications with safety instructions, and removes the product from the marketplace pending investigation.	
Actors:	System Admin, Pet Owner, Marketplace Vendor	
Related use cases:	Inventory Expiry Tracker, Ingredient Allergy Alert System, Multi-Channel Notification Escalator	
Stakeholders:	Pet Owners, Marketplace Vendors, Platform Administration, Regulatory Authorities	
Preconditions:	Recall information must be confirmed and entered by admin. Purchase history records must be available. Notification system must be operational.	
Postconditions:	All affected customers are notified. Recalled product is removed from marketplace. Recall event is logged for compliance.	
Flow of activities	Actor	System
	1. Admin logs into platform and enters recall details	1.1 System identifies all purchase records for recalled product
	2. Admin confirms recall and triggers notification	2.1 System sends immediate notifications to all affected owners
		2.2 System removes recalled product from marketplace and logs event
Exception conditions:	2.1 Notification delivery failure for some users — system logs failed deliveries and retries via alternative channels. 1.1 Product ID not found in system — admin must manually match purchase records.	

C - Service Provider & Logistics

Use Case: Dynamic Service Area Geofencing		
Use case name:	Dynamic Service Area Geofencing	
Scenario:	Service provider defining their active service coverage area	
Triggering event:	A service provider registers or updates their service area boundaries	
Brief description:	The system allows service providers to draw or define geofenced service areas on a map, restricting booking requests to those within the defined boundaries and alerting providers to out-of-area requests.	
Actors:	Service Provider, System Admin, Pet Owner	
Related use cases:	Real-time Walk GPS Tracker, Sitter Availability Conflict Resolver, Provider Certification Verification	
Stakeholders:	Service Providers, Pet Owners, Platform Administration	
Preconditions:	Service provider must have an active verified account. Mapping and geofencing module must be available. Provider must set at least one active service area.	
Postconditions:	Service area is saved and activated. Booking requests from outside area are filtered. Owner search results reflect provider availability by location.	
Flow of activities	Actor	System
	1. Provider logs in and navigates to service area settings	1.1 System loads mapping interface with current location
	2. Provider draws boundary on map and sets service radius	2.1 System validates area coverage and checks for conflicts
	3. Provider saves and activates geofence	3.1 System saves geofence and filters future booking requests accordingly
Exception conditions:	2.1 Defined area overlaps restricted zone — system highlights conflict and prompts correction. 3.1 Area too large for single provider — system warns and suggests splitting into sub-zones.	

Use Case: Tiered Service Pricing Engine	
Use case name:	Tiered Service Pricing Engine

Scenario:	Service provider setting pricing tiers for different services and pet types	
Triggering event:	Service provider creates or updates their service offerings with custom pricing	
Brief description:	The system enables service providers to define tiered pricing based on service type, pet size, duration, and time of day, automatically applying the correct price when owners book a service.	
Actors:	Service Provider, Pet Owner, System	
Related use cases:	Service Escrow Management, Multi-Vendor Commission Logic, Sitter Availability Conflict Resolver	
Stakeholders:	Service Providers, Pet Owners, Accounting Team	
Preconditions:	Service provider must be registered and verified. At least one service offering must be defined. Pricing rules must be configured by the provider.	
Postconditions:	Pricing tiers are saved and applied to booking interface. Owners see accurate price before confirming booking. Pricing is used for payment processing.	
Flow of activities	Actor	System
	1. Provider navigates to service pricing settings	1.1 System loads pricing configuration interface
	2. Provider defines pricing tiers by service, pet size, and time slot	2.1 System validates pricing entries and checks for conflicts
	3. Provider saves pricing configuration	3.1 System activates pricing and applies it to booking engine
Exception conditions:	2.1 Pricing conflict detected (overlapping tiers) — system highlights conflict and blocks save until resolved. 3.1 Pricing below platform minimum — system warns provider and requires adjustment.	

Use Case: Real-time Walk GPS Tracker	
Use case name:	Real-time Walk GPS Tracker
Scenario:	Pet owner tracking their dog's walk in real time
Triggering event:	Service provider (walker) starts a walk session for a booked service
Brief description:	The system activates GPS tracking when a walk session begins, sharing the live location with the pet owner and recording the route, distance, and duration for the session report.

Actors:	Service Provider (Walker), Pet Owner, System	
Related use cases:	Check-In/Check-Out QR Verification, Sitter-to-Owner Incident Reporting, Dynamic Service Area Geofencing	
Stakeholders:	Pet Owner, Service Provider, Platform Administration	
Preconditions:	Walk must be booked and confirmed. Walker must have the mobile app with location permissions enabled. Walk session must be officially started.	
Postconditions:	Walk route is recorded. Owner receives real-time location updates. Session summary with route map is shared post-walk.	
Flow of activities	Actor	System
	1. Walker taps 'Start Walk' in the app	1.1 System activates GPS tracking and begins broadcasting location to owner
	2. Walker completes the walk and taps 'End Walk'	2.1 System stops tracking and generates walk summary with route map
		2.2 System sends session report to owner and updates booking record
Exception conditions:	1.1 GPS signal unavailable — system notifies owner of tracking unavailability and logs manual check-in. 2.1 Walk session not ended by walker — system auto-closes session after inactivity timeout.	

Use Case: Sitter Availability Conflict Resolver	
Use case name:	Sitter Availability Conflict Resolver
Scenario:	System detecting and resolving booking conflicts for a pet sitter
Triggering event:	A new booking request is submitted for a sitter that overlaps with an existing booking
Brief description:	The system checks a sitter's availability calendar in real time when a booking is requested and automatically detects conflicts, blocking double-bookings and suggesting alternative available slots.
Actors:	Pet Owner, Service Provider (Sitter), System
Related use cases:	Tiered Service Pricing Engine, Dynamic Service Area Geofencing, Check-In/Check-Out QR Verification
Stakeholders:	Pet Owners, Service Providers, Platform Administration

Preconditions:	Sitter availability calendar must be configured and active. Sitter must be registered and verified. Booking system must be operational.	
Postconditions:	Conflict is detected and booking is blocked or re-routed. Owner is shown alternative available slots. Sitter's calendar remains accurate.	
Flow of activities	Actor	System
	1. Owner selects sitter and desired booking date/time	1.1 System checks sitter's availability calendar for conflicts
	2. Owner reviews alternative available slots if conflict exists	2.1 System blocks conflicting slot and returns available alternatives
		2.2 System confirms booking on owner's selected alternative slot
Exception conditions:	1.1 Sitter has no available slots in requested period — system suggests other qualified sitters in the area. 2.1 Simultaneous booking conflict — system applies first-come-first-served logic and notifies second requester.	

Use Case: Provider Certification Verification		
Use case name:	Provider Certification Verification	
Scenario:	Platform verifying the certifications and qualifications of a service provider	
Triggering event:	A service provider submits or updates their professional certification documents	
Brief description:	The system processes submitted certification documents, verifies their authenticity, and updates the provider's verified status badge on the platform.	
Actors:	Service Provider, System Admin, Verification Engine	
Related use cases:	Role-Based Access Control, User Identity Verification (KYC), Dispute Mediation Workflow	
Stakeholders:	Service Providers, Pet Owners, Platform Administration, Legal/Compliance Team	
Preconditions:	Provider must have a registered account. Certification documents must be uploaded in accepted formats. Verification rules must be configured by admin.	
Postconditions:	Certification is verified or rejected with reason. Provider's status badge is updated. Verified provider gains access to premium service tiers.	
Flow of activities	Actor	System

	1. Provider uploads certification documents	1.1 System receives documents and queues for verification
	2. Admin reviews verification request	2.1 System cross-references document details with certification database
		2.2 System updates provider certification status and badge
Exception conditions:	2.1 Certification cannot be verified automatically — system flags for manual admin review. 1.1 Unsupported document format uploaded — system rejects upload and specifies accepted formats.	

Use Case: Sitter-to-Owner Incident Reporting		
Use case name:	Sitter-to-Owner Incident Reporting	
Scenario:	Sitter reporting an incident that occurred during a pet care session	
Triggering event:	An incident occurs during a pet sitting or walking session	
Brief description:	The system provides a structured incident reporting form for service providers, immediately notifying the pet owner, logging the report, and escalating to admin for serious incidents.	
Actors:	Service Provider (Sitter/Walker), Pet Owner, System Admin	
Related use cases:	Real-time Walk GPS Tracker, Check-In/Check-Out QR Verification, Dispute Mediation Workflow	
Stakeholders:	Service Provider, Pet Owner, Platform Administration, Legal/Insurance Team	
Preconditions:	Active booking session must exist. Service provider must be authenticated. Incident reporting module must be operational.	
Postconditions:	Incident report is created and stored. Owner is immediately notified. Admin is alerted for serious incidents. Report is linked to booking record.	
Flow of activities	Actor	System
	1. Sitter opens incident report form during active session	1.1 System loads incident form with active booking details pre-filled
	2. Sitter describes incident, selects severity, and uploads photo if available	2.1 System validates form completeness

	3. Sitter submits report	3.1 System submits report, notifies owner, and escalates to admin if severity is high
Exception conditions:	3.1 Notification delivery to owner fails — system retries and logs failure for admin follow-up. 2.1 Session has expired before report submission — system allows report within 2-hour post-session window.	

Use Case: Service Escrow Management		
Use case name:	Service Escrow Management	
Scenario:	Platform holding payment in escrow during a service booking	
Triggering event:	Owner confirms and pays for a booked service	
Brief description:	The system holds the service payment in escrow upon booking confirmation, releasing funds to the service provider only after the owner confirms service completion or a defined time period elapses without a dispute.	
Actors:	Pet Owner, Service Provider, System, Payment Gateway	
Related use cases:	Tiered Service Pricing Engine, Dispute Mediation Workflow, Multi-Vendor Commission Logic	
Stakeholders:	Pet Owners, Service Providers, Accounting/Finance Team, Legal Team	
Preconditions:	Booking must be confirmed by both owner and provider. Payment method must be valid. Escrow rules must be configured.	
Postconditions:	Payment is held in escrow during service. Funds are released to provider after successful completion. Escrow transaction is logged for accounting.	
Flow of activities	Actor	System
	1. Owner confirms booking and makes payment	1.1 System charges owner and holds funds in escrow
	2. Owner confirms service completion after session	2.1 System releases held funds to provider upon confirmation
		2.2 System deducts platform commission and updates accounting records
Exception conditions:	2.1 Owner raises dispute before release — system freezes escrow and triggers Dispute Mediation Workflow.	

	1.1 Payment authorization fails — system cancels booking and notifies owner to retry.
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Use Case: Check-In/Check-Out QR Verification		
Use case name:	Check-In/Check-Out QR Verification	
Scenario:	Service provider verifying pet drop-off and pick-up using QR codes	
Triggering event:	A pet is dropped off or picked up for a booked service	
Brief description:	The system generates a unique QR code for each booking, which the service provider scans at check-in and check-out to officially record the pet's arrival and departure, notifying the owner at each event.	
Actors:	Service Provider, Pet Owner, System	
Related use cases:	Sitter Availability Conflict Resolver, Real-time Walk GPS Tracker, Service Escrow Management	
Stakeholders:	Pet Owners, Service Providers, Platform Administration	
Preconditions:	Booking must be confirmed and active. QR code must be generated and sent to owner. Service provider must have QR scanner access in the app.	
Postconditions:	Check-in or check-out event is recorded with timestamp. Owner is notified. Session status is updated in booking record.	
Flow of activities	Actor	System
	1. Owner presents QR code at drop-off	1.1 System generates unique QR code and sends to owner upon booking confirmation
	2. Provider scans QR code using app	2.1 System records check-in with timestamp and notifies owner
	3. Provider scans QR code again at pick-up	3.1 System records check-out and triggers service completion notification
Exception conditions:	2.1 QR code scan fails — system allows manual code entry as fallback. 3.1 Check-out not recorded within session window — system alerts admin for follow-up.	

Use Case: Behavioral Profile Sharing

Use case name:	Behavioral Profile Sharing	
Scenario:	Owner sharing pet behavioral profile with a new service provider	
Triggering event:	A booking is confirmed with a service provider who has not previously cared for the pet	
Brief description:	The system automatically shares the pet's behavioral profile, including temperament, known triggers, special handling instructions, and medical alerts, with the assigned service provider upon booking confirmation.	
Actors:	Pet Owner, Service Provider, System	
Related use cases:	Veterinary Referrals Workflow, Sitter-to-Owner Incident Reporting, Check-In/Check-Out QR Verification	
Stakeholders:	Pet Owner, Service Provider, Veterinary Clinic	
Preconditions:	Pet behavioral profile must be completed by owner or vet. Booking must be confirmed. Owner must grant profile sharing consent.	
Postconditions:	Behavioral profile is shared securely with provider. Provider receives notification with profile summary. Sharing event is logged.	
Flow of activities	Actor	System
	1. Owner completes or updates pet behavioral profile	1.1 System saves behavioral profile to pet record
	2. Owner confirms booking and grants sharing consent	2.1 System shares relevant behavioral data with assigned provider
		2.2 System notifies provider and logs sharing event
Exception conditions:	2.1 Owner denies sharing consent — system notifies provider of limited pet information availability. 1.1 Behavioral profile incomplete — system prompts owner to complete key fields before confirming booking.	

D - System Administration & Security

Use Case: Role-Based Access Control (RBAC)		
Use case name:	Role-Based Access Control (RBAC)	
Scenario:	System administrator assigning access roles to platform users	
Triggering event:	A new user registers or an existing user's role requires updating	
Brief description:	The system enforces access control by assigning defined roles (e.g., Owner, Vet, Provider, Admin) to users, restricting feature access based on role permissions and logging all access-level changes.	
Actors:	System Admin, Platform Users, System	
Related use cases:	User Identity Verification (KYC), Data Privacy Audit Logger, Automated Account Suspension Logic	
Stakeholders:	All Platform Users, IT/Security Team, Compliance Team	
Preconditions:	User must have a registered account. Role definitions and permissions must be configured. Admin must be authenticated.	
Postconditions:	User role is assigned or updated. Access permissions are applied immediately. Role change is logged in the audit trail.	
Flow of activities	Actor	System
	1. Admin navigates to user management panel	1.1 System loads user list with current roles
	2. Admin selects user and assigns or updates role	2.1 System validates role assignment against defined permission matrix
		2.2 System applies new role permissions and logs change in audit trail
Exception conditions:	2.1 Role conflict detected — system blocks assignment and alerts admin. 2.2 Unauthorized admin action — system denies change and triggers security alert.	

Use Case: Emergency Lost Pet Broadcast	
Use case name:	Emergency Lost Pet Broadcast
Scenario:	Pet owner reporting a lost pet and broadcasting an emergency alert

Triggering event:	Owner marks their pet as lost through the platform	
Brief description:	The system broadcasts an emergency lost pet alert to nearby registered platform users, service providers, and veterinary clinics within a configurable radius, sharing the pet's profile, last known location, and owner contact details.	
Actors:	Pet Owner, System, Nearby Platform Users, Service Providers	
Related use cases:	Real-time Walk GPS Tracker, Multi-Channel Notification Escalator, Behavioral Profile Sharing	
Stakeholders:	Pet Owner, Nearby Platform Users, Service Providers, Veterinary Clinics, Platform Administration	
Preconditions:	Pet profile must be complete with photo and description. Owner must be authenticated. Geolocation services must be active. Notification system must be operational.	
Postconditions:	Lost pet alert is broadcast to all nearby users within the defined radius. Owner receives confirmation of broadcast. Alert remains active until owner marks pet as found.	
Flow of activities	Actor	System
	1. Owner opens app and selects 'Report Lost Pet'	1.1 System loads pet profile and prompts owner to confirm details
	2. Owner confirms last known location and pet details	2.1 System captures last known location via GPS or manual entry
	3. Owner activates the broadcast	3.1 System broadcasts alert to all registered users and providers within defined radius
		3.2 System sends alert via push notification, email, and SMS
Exception conditions:	3.1 No registered users found within broadcast radius — system expands radius and retries. 2.1 Pet profile photo missing — system prompts owner to upload photo before broadcasting.	

Use Case: Multi-Channel Notification Escalator	
Use case name:	Multi-Channel Notification Escalator
Scenario:	System escalating an unacknowledged notification through multiple channels
Triggering event:	A critical notification is sent but not acknowledged by the recipient within a defined time window

Brief description:	The system monitors notification acknowledgment and automatically escalates unacknowledged critical alerts through secondary and tertiary channels (in-app, email, SMS, phone) until confirmation is received.	
Actors:	System, Pet Owner, Service Provider, Admin	
Related use cases:	Emergency Red-Flag Logic, Product Recall Notification System, Automated Account Suspension Logic	
Stakeholders:	All Platform Users, IT Team, Platform Administration	
Preconditions:	Notification must be classified as critical. Escalation rules and timing windows must be configured. User must have multiple contact channels registered.	
Postconditions:	Notification is delivered and acknowledged via at least one channel. Escalation chain is logged. Admin is notified if all channels fail.	
Flow of activities	Actor	System
	1. User receives initial notification (in-app)	1.1 System sends notification and starts acknowledgment timer
	2. User acknowledges notification via any channel	2.1 System detects non-acknowledgment and escalates to next channel after timeout
		2.2 System records successful delivery and acknowledgment
Exception conditions:	2.1 All channels fail — system flags user account for admin manual outreach. 1.1 User contact information incomplete — system skips unavailable channel and proceeds to next.	

Use Case: Dispute Mediation Workflow	
Use case name:	Dispute Mediation Workflow
Scenario:	Owner or service provider raising a formal dispute about a transaction or service
Triggering event:	Owner or provider submits a dispute claim through the platform
Brief description:	The system captures dispute details, notifies both parties, assigns a mediation case, collects evidence from both sides, and facilitates resolution through admin review or automated settlement rules.
Actors:	Pet Owner, Service Provider, System Admin, Mediation System

Related use cases:	Service Escrow Management, Automated Account Suspension Logic, Data Privacy Audit Logger	
Stakeholders:	Pet Owners, Service Providers, Platform Administration, Legal Team	
Preconditions:	A completed or active booking must exist. Dispute must be submitted within the allowed dispute window. Both parties must have active accounts.	
Postconditions:	Dispute case is opened and both parties are notified. Evidence is collected and stored. Resolution is applied and parties are informed. Escrow is released or refunded accordingly.	
Flow of activities	Actor	System
	1. Owner or provider submits dispute claim with evidence	1.1 System opens dispute case and notifies both parties
	2. Opposing party responds to claim	2.1 System collects and stores evidence from both parties
	3. Admin reviews and issues resolution	3.1 System applies resolution decision and adjusts escrow/payment accordingly
Exception conditions:	2.1 Opposing party does not respond within deadline — system applies default resolution rules. 3.1 Admin cannot reach resolution — system escalates to legal team and freezes relevant funds.	

Use Case: Data Privacy Audit Logger	
Use case name:	Data Privacy Audit Logger
Scenario:	System automatically logging all data access and modification events for audit purposes
Triggering event:	A user accesses, modifies, exports, or deletes sensitive data on the platform
Brief description:	The system records every access and modification event related to sensitive data (medical records, personal information, financial data) with timestamps, user identity, and action type, making logs available for compliance audits.
Actors:	System, System Admin, Compliance Officer
Related use cases:	Role-Based Access Control, User Identity Verification (KYC), Automated Account Suspension Logic
Stakeholders:	All Platform Users, Legal/Compliance Team, IT Security Team, Regulatory Authorities

Preconditions:	Audit logging module must be enabled. Sensitive data categories must be defined. System time synchronization must be accurate.	
Postconditions:	All qualifying events are logged with full details. Logs are tamper-proof and time-stamped. Compliance reports can be generated from audit trail.	
Flow of activities	Actor	System
	1. User performs an action involving sensitive data	1.1 System intercepts action and generates audit log entry
	2. Compliance officer requests audit report	2.1 System compiles and exports audit report for specified period
		2.2 System provides read-only access to filtered log entries
Exception conditions:	1.1 Logging service unavailable — system blocks action and alerts admin to prevent unlogged data access. 2.1 Report generation failure — system provides partial data and logs error for IT review.	

Use Case: Automated Account Suspension Logic		
Use case name:	Automated Account Suspension Logic	
Scenario:	System automatically suspending an account that violates platform rules	
Triggering event:	A user account triggers one or more predefined suspension rules (e.g., repeated policy violations, failed payments, suspicious activity)	
Brief description:	The system evaluates user behavior against suspension rules, automatically suspending violating accounts, notifying the user, and providing an appeal pathway through admin review.	
Actors:	System, System Admin, Platform User	
Related use cases:	Role-Based Access Control, Data Privacy Audit Logger, Dispute Mediation Workflow	
Stakeholders:	All Platform Users, Platform Administration, Legal Team	
Preconditions:	Suspension rules must be defined and activated by admin. User account must be active. Monitoring engine must be running.	
Postconditions:	Non-compliant account is suspended. User is notified with reason. Account data is preserved. Appeal process is made available.	
Flow of activities	Actor	System

	1. User performs action that triggers suspension rule	1.1 System detects rule violation and evaluates suspension threshold
	2. User receives suspension notice and reviews appeal options	2.1 System suspends account and sends automated notification with reason
		2.2 System logs suspension event and presents appeal form to user
Exception conditions:	1.1 False positive detected during appeal — admin manually reinstates account and adjusts rule sensitivity. 2.1 User does not acknowledge suspension notice — system escalates to admin for manual follow-up.	

Use Case: System-Wide Health Alerts		
Use case name:	System-Wide Health Alerts	
Scenario:	System monitoring platform infrastructure and alerting admins to issues	
Triggering event:	A system component experiences degraded performance or failure beyond defined thresholds	
Brief description:	The system continuously monitors infrastructure health metrics (server load, API response times, database performance) and automatically sends alerts to admins when thresholds are breached, with suggested remediation steps.	
Actors:	System Monitoring Engine, System Admin, DevOps Team	
Related use cases:	Multi-Channel Notification Escalator, Automated Account Suspension Logic, Data Privacy Audit Logger	
Stakeholders:	Platform Administration, IT/DevOps Team, All Platform Users	
Preconditions:	Monitoring module must be configured and active. Health check thresholds must be defined. Admin notification channels must be set up.	
Postconditions:	Health alert is sent to admin and DevOps team. Affected component is flagged in the monitoring dashboard. Incident is logged for post-mortem review.	
Flow of activities	Actor	System
	1. Admin configures health monitoring thresholds	1.1 System activates continuous health monitoring for all components

	2. Admin receives alert and investigates issue	2.1 System detects threshold breach and sends alert with component details
		2.2 System logs incident and provides suggested remediation checklist
Exception conditions:	2.1 Monitoring engine itself experiences failure — backup watchdog triggers alert via alternative channel. 1.1 Threshold set too low causing false alarms — admin recalibrates threshold and system resumes normal monitoring.	

Use Case: User Identity Verification (KYC)		
Use case name:	User Identity Verification (KYC)	
Scenario:	Platform verifying the identity of a new user or service provider	
Triggering event:	A new user registers as a service provider or requests elevated account privileges	
Brief description:	The system collects identity documents and biometric data from the user, processes them through a KYC verification engine, and updates the user's verified status to unlock appropriate platform features.	
Actors:	Platform User, KYC Verification Engine, System Admin	
Related use cases:	Role-Based Access Control, Provider Certification Verification, Automated Account Suspension Logic	
Stakeholders:	Platform Users, Legal/Compliance Team, IT Security Team, Regulatory Authorities	
Preconditions:	User must have created a basic account. KYC verification module must be active and integrated. Accepted document types must be defined.	
Postconditions:	Identity is verified or rejected with reason. Verified status is updated on user profile. KYC result is stored for compliance. Unverified users have restricted access.	
Flow of activities	Actor	System
	1. User submits identity documents and selfie via app	1.1 System uploads documents to KYC engine for processing
	2. User awaits verification outcome	2.1 System receives verification result and updates user status

		2.2 System notifies user of verification outcome and unlocked features
Exception conditions:	2.1 Document rejected (blurry, expired, or mismatched) — system notifies user with specific rejection reason and allows resubmission. 1.1 KYC service unavailable — system queues submission and notifies user of expected delay.	